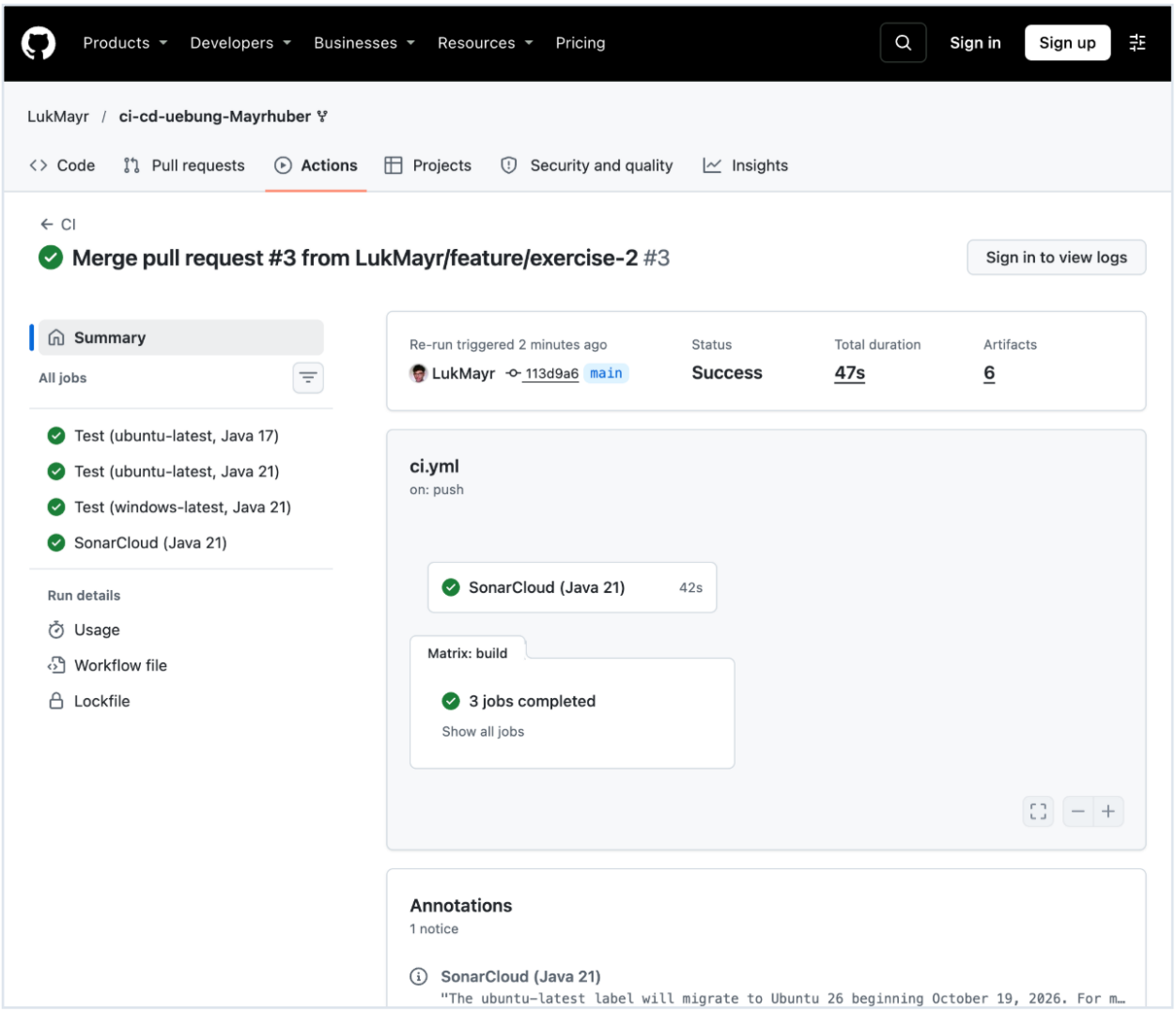
CI matrix and automated tests

UE02 | GitHub Actions + Maven/Surefire + JaCoCo + SonarCloud

The workflow builds and tests each push to **main**. Pull requests run the matrix when sources, the POM or workflow change. The verified main commit is **113d9a6**; the corrected SonarCloud job completed on rerun.



Matrix: Ubuntu/Java 17, Ubuntu/Java 21, Windows/Java 21. Windows/Java 17 is excluded to reduce runner time while retaining both operating systems and both Java versions. That specific combination remains untested.

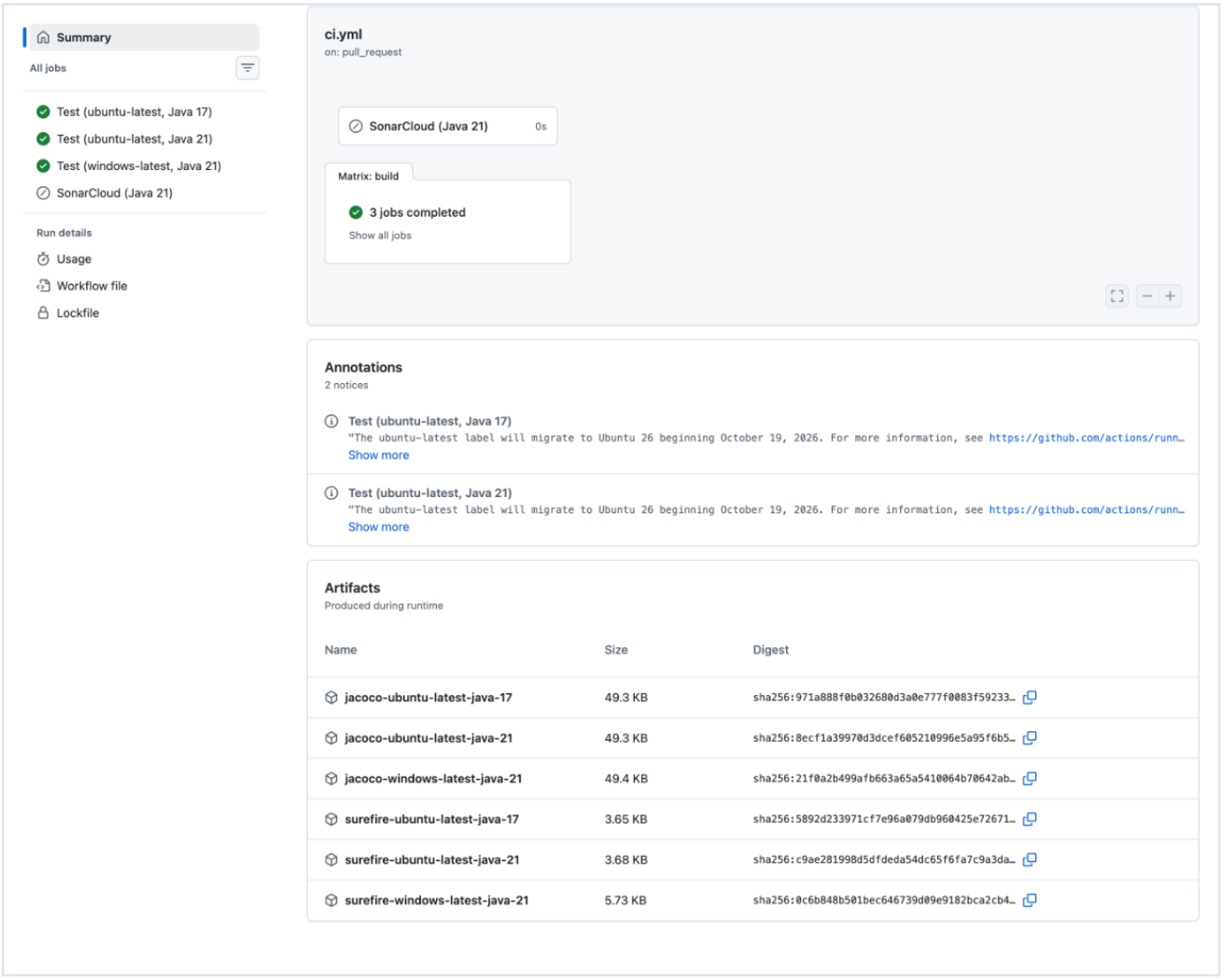
Execution: Temurin JDKs, Maven `clean verify`, 16 tests per variant, no skipped tests. Surefire reports upload even on test failure. PR concurrency cancels obsolete runs; main commits have separate groups.

Main CI run #3 (successful rerun)

Repository and workflow badge

Test and coverage artifacts

One clearly named report set per matrix combination



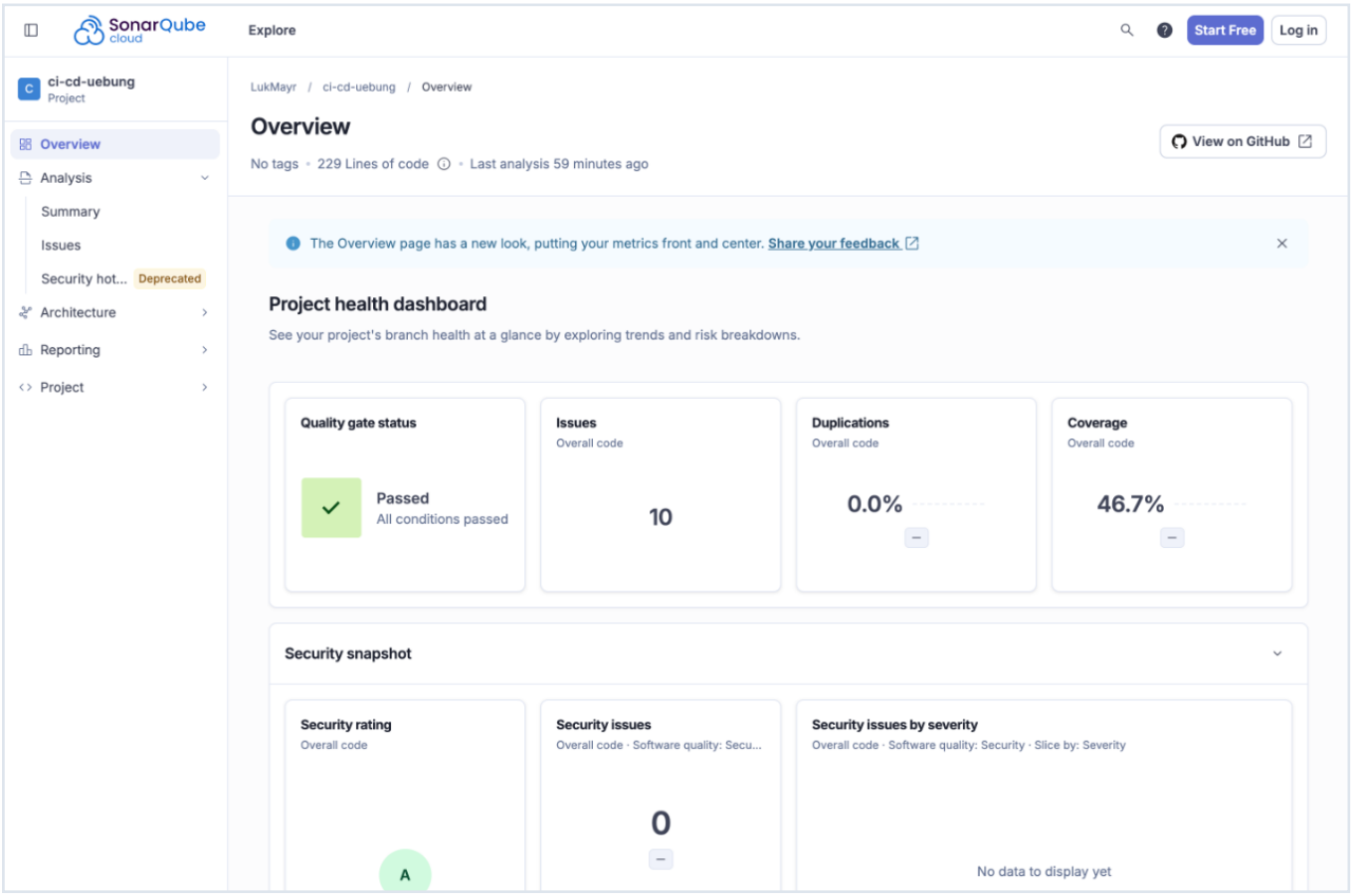
The screenshot from PR validation run #2 shows the three Surefire report archives and three JaCoCo archives. Each name identifies the operating system and JDK. The same six artifact types are present in the verified main run. Coverage archives contain HTML and XML; CI imports `target/site/jacoco/jacoco.xml` into SonarCloud.

Surefire artifacts contain the JUnit XML and text results. No tests are ignored to make the pipeline pass. The original baseline had 8 tests with one intentional palindrome failure; the corrected suite has 16 passing tests.

Artifact screenshot source: PR validation run #2

SonarCloud on main

CI analysis, imported coverage and a passing Quality Gate



Verified results: Quality Gate passed. Overall SonarCloud coverage is **46.7%**. The project has **0 bugs**, **0 vulnerabilities** and **10 remaining code smells**. Existing unrelated findings remain visible; the two selected findings are resolved.

Organization: **lukmayr**. Project key: **LukMayr_ci-cd-uebung-Mayrhuber**. The public keys and pinned Maven scanner are in the POM. **SONAR_TOKEN** is an Actions repository secret; Automatic Analysis is disabled. The workflow waits for the gate and fails when it fails.

SonarCloud project dashboard

Two confirmed findings fixed

Baseline findings saved before the confirming CI analysis

1. java:S4973 - incorrect string comparison
TextUtils.java, original line 8; issue AaGr8PdCU3o20iwK5k0w. Reference equality rejected the palindrome "Anna". Content comparison fixes the bug and avoids locale-dependent lowercase conversion.

```
// Before
return input.toLowerCase() == reversed.toLowerCase();

// After
return input.equalsIgnoreCase(reversed);
```

Regression checks: Anna, empty input, one character, abba and RaceCar return true; null, hello, ab and Anna! return false.

2. java:S4144 - identical method implementations
Calculator.java, original line 42; issue AaGr8PdBU3o20iwK5k0v. Both sum methods contained the same null-filtering loop. Delegating addAll to sumUp removes the duplicate implementation without changing behavior.

```
// Before: addAll repeated the loop from sumUp
int s = 0;
for (Integer n : nums) {
    if (n != null) { s += n; }
}
return s;

// After: addAll delegates to the shared implementation
return sumUp(nums);
```

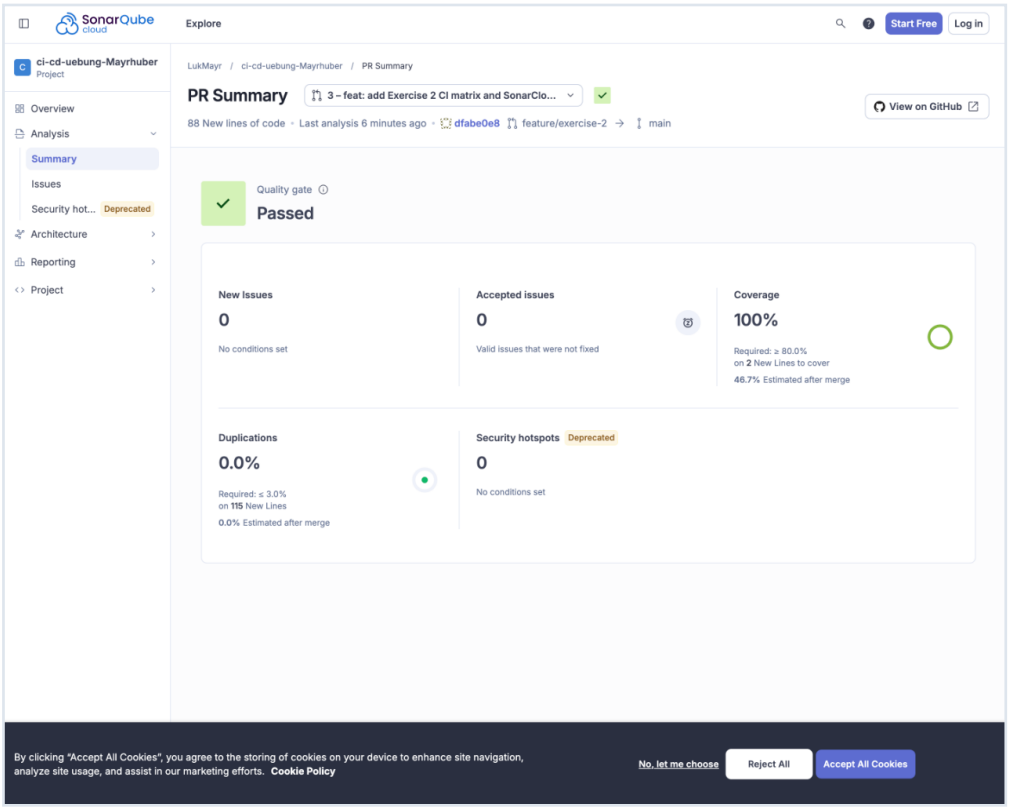
The strengthened tests assert concrete totals for both methods: 10 for positive values with null entries and -150 for negative values with null entries. Main analysis confirms both rule findings are absent from the open-issue set.

Merged implementation and SonarCloud PR decoration

Problems, solutions and evidence

Validation and remaining submission steps

Local JDK: Maven defaulted to Java 27. Validation explicitly used installed Java 17 and 21; both clean builds passed.
GitHub access: the CLI token was invalid. Git operations used SSH, and PR/CI operations used the connected GitHub integration.
Secret scope: the first main SonarCloud job received an empty token. The token was corrected under Actions repository secrets and the failed job was rerun.
Analysis mode: Automatic Analysis was disabled so CI owns analysis and imports coverage.



The supplementary local PR analysis passed with zero new issues, 100% coverage on the changed code, and 0% duplication on new code. The hosted main analysis separately verifies the required push-to-main workflow.

Submission: upload this PDF and the repository link to eLearning. The exercise sheet lists 6 November 2025, while the repository course material uses 2026; confirm the current course deadline. No credentials are included in the report or repository.

PR analysis dashboard